Aryan Ajith Nair
Dr. Bodwin
STAT 431
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Checkpoint 1 (Data = Measles Cases)

1. Describe Context:

This dataset explores measles and rubella cases across the world. The data comes from the World Health Organisation and covers data from 2012 to 2025. The number of cases of measles and rubella is only available by July of each year. The full r dataset with the month and year information is stored in tt_data. The month and year are stored in separate data objects that are tibbles.

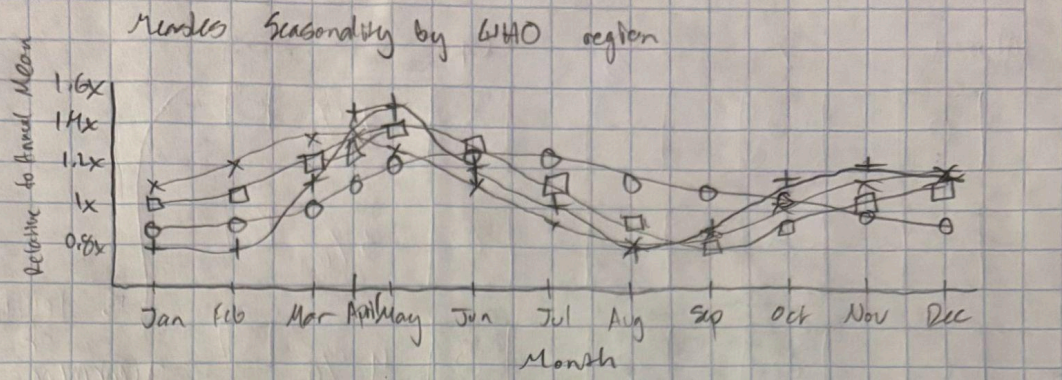
2. Cleaning:

For the cases by month data, we use the clean_names function from the janitor package which converts all column names to be all lowercase with spaces and special characters replaced with underscores. This makes the column names consistently formatted so they are easy to work with. With the mutate function, the non-string columns are converted into numeric data in case they weren't already. For the cases by year data, the first row was blank so we had to remove it and use the 2nd row as the column headers. The column names were manually renamed as there were placeholder names before. The slice(-1) function is what removes the first blank row.

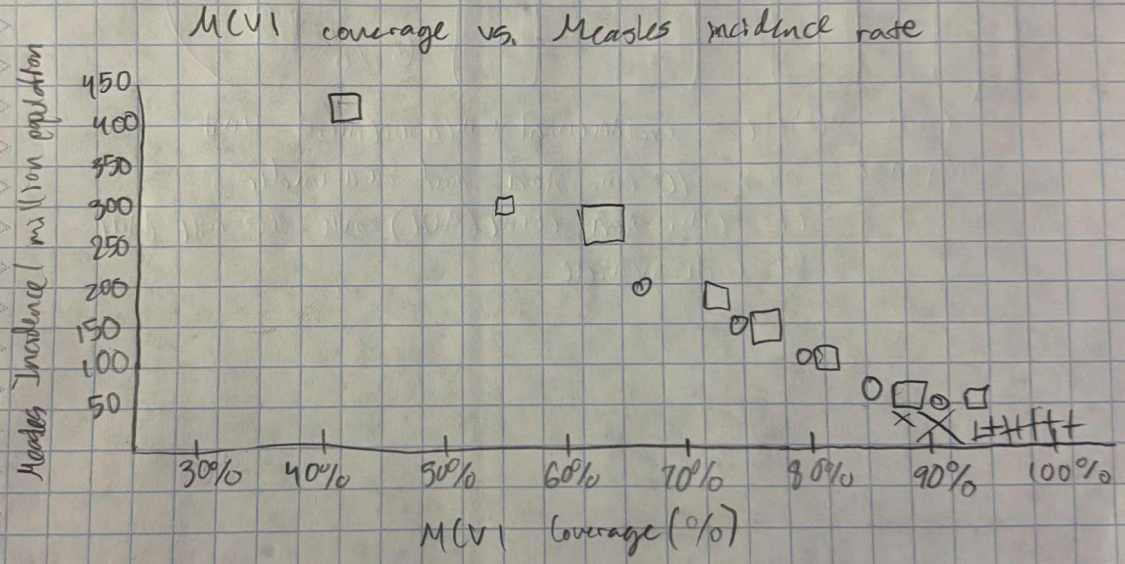
3. 2 Research Questions possible with given data:
    a. Has measles case confirmation quality improved over time?
    b. Does the seasonality of measles outbreaks differ by region?

4. 2 Research Questions possible with supplemental data:
    a. Does measles incidence correlate with vaccination coverage gaps?
    b. Does political instability or conflict predict outbreak magnitude?

5. Sketch of 2 Visualizations possible to address the Research Questions:



X = W. Pacific □ = Africa ○ = E. Mediterranean + = Europe



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